

Hot and Cold Lunch Box for Multifunctional Food management

Field of Invention

The present invention relates to novel design Hot and Cold Lunch Box is mainly used to keep food hot or cold based on the food.

Background

This box is to serve a purpose in food management, to keep the freshness state of food products by keeping them hot or cold depending on the food item. The gadget has a Induction, which is essential for heating of the food and mini Compressor to keep the food cool with switch over mechanism. This capability is critical for users who want to transport food with its best quality. Using this gadget, users can heat or cool the food kept inside accordingly.

KR102289148B1 displays “Lunchbox”, this prior art relates to a lunch box for storage, transport, and heating of food, in particular cooked food. It should be easy to handle. The lunch box according to the invention consists of a cover (1) and a body (2), respectively, and a locking device is attached to the vertical part so that it can be opened and closed after being placed overlapping each other. The body 2 has one inner container with a cylindrical and L-shaped rim. In this inner container, a heat insulator and a heating element are arranged. A groove is formed in the rim of the inner container to receive the sealing element.

US20210347552A1 discloses “Food container using heating element” a food container using a heating element, which container enhances convenience since a user can simply inject water toward the heating element by opening an inflow passage of an outer container without removing an inner container, and can keep the heating element stable in a simplified structure since a space containing the heating element inside the outer container and a space containing water are formed to be divided by a protrusion jaw but to communicate with each other.

While the cited references disclose different types of lunch box, typically made of steel or plastic, lacked the ability to keep food warm or cool respectively for an extended period. This often resulted in kids having cold or lukewarm meals during lunchtime, affect the overall eating experience and nutritional quality of the food.

This novel Hot and Cold Lunch Box, specifically tailored to keep food fresh, serves as a tool in safeguarding food quality during transportation and storage. By integrating advanced sensing technologies, including temperature, humidity, gas emissions, and microbial activity sensors, this device offers comprehensive monitoring capabilities within hot or cold boxes. Existing technologies predominantly focus on insulation mechanisms tailored to either hot or cold food storage / transportation, with minimal consideration for accommodating both requirements concurrently. Therefore, in light of the foregoing discussion, there exists a need to overcome the aforementioned drawbacks.

Patent Claims

- The principal object of the present invention is to overcome the shortcomings, drawbacks, limitations, and shortcomings associated with the existing arts.
- The present disclosure is to provide a lunch box solution that can heat or cool food ensuring that people have access to warm or cool and fresh meals.
- The present disclosure is to offer a practical and convenient solution for people to ensure that they have access to warm or cold meals, even in situations where there are no appliances available to keep the food warm or cold.
- To introduce a significant advancement in lunch box technology, by providing means to keep the food at its best temperature.
- To enhance the portability and usability of lunch box by incorporating a heating and cooling mechanism, making it suitable for various settings such as school, picnics, or travel to transport the food as required.

Summary of the Invention

The objective of the present disclosure is to provide a versatile solution for maintaining both hot and cold food in the box simultaneously within the containers, thereby enhancing user convenience and satisfaction of the food consumption.

Brief description of the drawings

Embodiments of the present disclosure will now be described. FIG. 1 is a schematically illustration of a block diagram of the lunch box, in accordance with an embodiment of the present disclosure; FIG. 2 is a schematically illustration of a diagram of the lunch box designed to maintain both hot and cold beverages simultaneously, in accordance with an embodiment of the present disclosure;

Detailed description

The following detailed description illustrates embodiments of the present disclosure and ways in which they can be implemented. The description set forth below in connection with the appended drawings is intended as a description of certain embodiments of the system and is not intended to represent the only forms that may be developed or utilized. The description sets forth the various structures and/or functions in connection with the illustrated embodiments.

While the disclosure is susceptible to various modifications and alternative forms, specific embodiment thereof has been shown by way of example in the drawings 10 and will be described in detail below. It should be understood, however, that it is not intended to limit the disclosure to the particular forms disclosed, but on the contrary, the disclosure is to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure.

FIG. 1 is a schematically illustration of a block diagram of a bottle, in accordance with an embodiment of the present disclosure. With reference to FIG.1, there is shown a block diagram 100 of the lunch box 101 comprising the containers with stainless steel, 102 induction coils to heat the containers with PU

insulation filling and a vacuum layer, 103 compressors to cool the containers, 104 switch for turning the heat and cold for the containers, 105 the power socket to heat or cool the food in the container.

FIG. 2 is a schematically illustration of a diagram of the lunch box designed to maintain both hot and cold food items simultaneously, in accordance with an embodiment of the present disclosure. With reference to FIG.2, there is shown a diagram that includes the lunch box designed to maintain both hot and cold food simultaneously. FIG 2 shows the different views of the lunch box to have better understanding of the embodiment.

Claims

I/We Claim

1. The Hot and Cold Lunch Box (100) configured to heat and cool the food in the containers (101) comprising of stainless steel, PU insulation filling, and a vacuum layer and also configured to insulate and cool food. A direct locking mechanism ensuring a secure seal for the box
2. The Hot and Cold Lunch Box (100) as claimed in claim 1, wherein the containers (101) has a capacity of 500ml.
3. The Hot and Cold Lunch Box (100) as claimed in claim 1, wherein the stainless steel containers (101) is insulated to retain the heat for longer time.
4. The Hot and Cold Lunch Box (100) as claimed in claim 1, wherein the PU insulation filling of the condenser is designed to enhance thermal insulation for cooling the containers.
5. The Hot and Cold Lunch Box (100) as claimed in claim 1, wherein the switchover mechanism allows for easy access to either switch the hot or cold compartment with minimal effort.
6. The Hot and Cold Lunch Box (100) as claimed in claim 1, wherein the direct locking mechanism ensures a leak-proof seal.
7. The Hot and Cold Lunch Box (100) as claimed in claim 1, wherein the overall design and construction of the lunch box prioritize user convenience, durability, and efficient temperature maintenance for both hot and cold food items.

Drawings

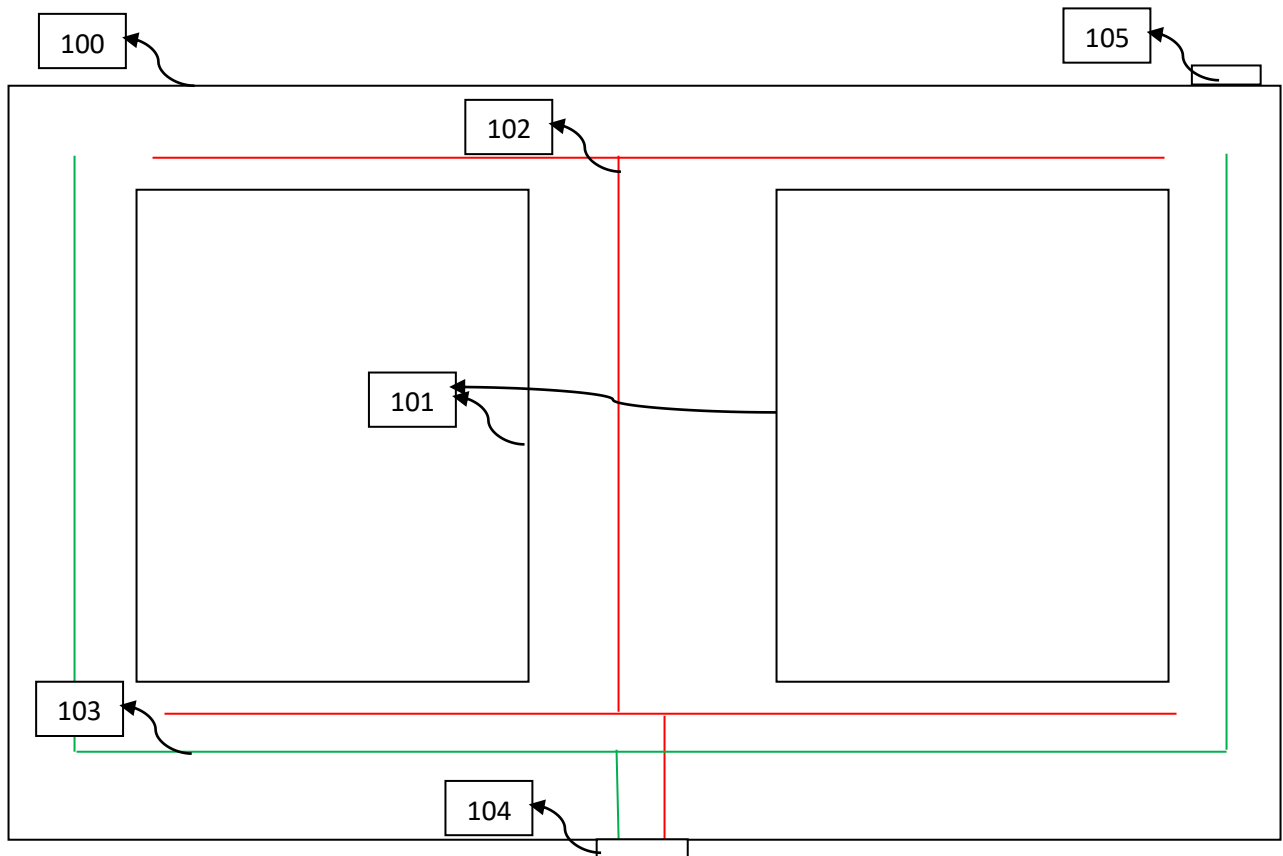


Fig 1

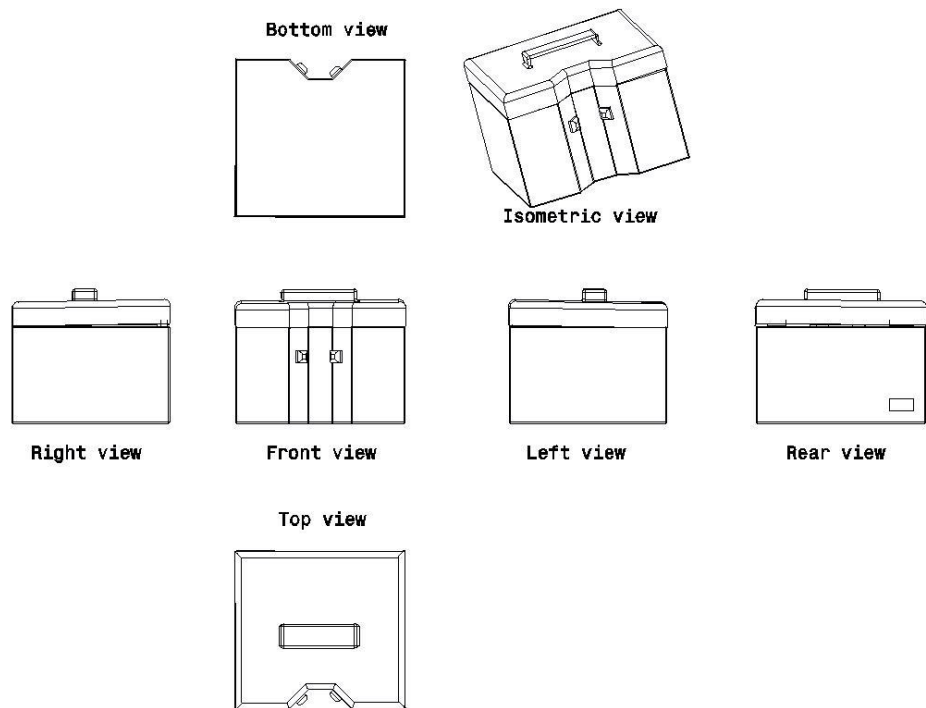


Fig 2

Abstract

The present invention relates to battery powered lunch box apparatus with a lunch box body consists of two main compartments each designed to hold different food items, the heating mechanism includes induction heating element covering side and base of the lunch box, the cooling mechanism includes compressor covering front and back of the lunch box, a user-friendly control panel includes an hot or cold switch, with inbuilt led indicator light to display the operation being used, a thermostat sensor to auto cut the heating element, power socket to power the device. The layers of the box is comprising a triple-layered construction comprising stainless steel, PU insulation filling, and a vacuum layer, a section configured to insulate cooling mechanism, a switch or press mechanism for accessing either temperature zone and a direct locking mechanism ensuring a secure seal between the induction and compressor.